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Digital Payment Technology and Financial Inclusion Among Rural Women: Evidence From Makueni County, Kenya

Nicholas Makusa*

School of Business and Economics, Catholic University of Eastern Africa, 62157-00200 Nairobi, Kenya

* Correspondence: Nicholas Makusa (nicholasmakusa@gmail.com)

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Abstract: Digital payment technologies are reshaping access to financial services in developing economies, yet rural women continue to face constraints related to connectivity, affordability, digital skills, and access to formal financial institutions. This study investigates the relationship between digital payment technology and financial inclusion among rural women in Makueni County, Kenya, with particular attention to accessibility, convenience, and participation in financial activities. A mixed-methods cross-sectional design was adopted. Quantitative data were collected from 372 rural women through questionnaires, while qualitative evidence was obtained from interviews with 30 digital financial service agents. Descriptive statistics, Pearson correlation, and multiple regression were used to analyze the quantitative data, and interview responses were examined thematically. The results showed a high level of acceptance and use of digital payment services, with a mean (M) of 4.17 and a standard deviation (SD) of 0.75, particularly in reducing the need to travel to banking institutions (M = 4.31, SD = 0.65) and improving local access to financial services (M = 4.28, SD = 0.71). Digital payment solutions were strongly associated with financial inclusion, with a Pearson correlation coefficient (r) of 0.731 and a probability value (p) of less than 0.01. In the regression model, which also included digital financing and digital advisory services, digital payment solutions showed the largest positive association with financial inclusion, with an unstandardized regression coefficient (B) of 0.382 ($p < 0.001$), while the three predictors jointly explained 70.9% of its variance. The qualitative findings further identified network reliability, transaction costs, fraud concerns, and digital skills as practical conditions shaping the use of digital payment services. The findings indicate that digital payment technology is closely associated with rural women's participation in formal financial systems, although the cross-sectional design does not permit causal inference. The study provides context-specific evidence on how accessible and usable digital financial technologies can support women's financial participation and small-scale economic activity in rural communities, with implications for technology providers, financial institutions, and policymakers.

Keywords: Digital payment technology; Financial inclusion; Rural women; Digital financial services; Technology adoption; Kenya

1. Introduction

Digital payment solutions have emerged as one of the most transformative innovations in advancing financial inclusion across developing countries. These solutions, which include mobile money services, digital wallets, electronic fund transfers, and mobile banking platforms, have significantly improved access to financial services, particularly among underserved populations. According to the International Telecommunication Union (International Telecommunication Union, 2022), digital financial services are broadly defined as financial services delivered through digital technologies that facilitate payments, savings, transfers, and other financial transactions. Globally, digital payment solutions have become instrumental in addressing barriers such as geographical distance, high transaction costs, and limited access to formal banking institutions, especially among rural women. However, the extent to which women can benefit from these services depends critically on their access to mobile technology. According to the Global System for Mobile Communications Association (GSMA) Mobile Gender Gap Report 2024, while Kenya has nearly closed the gender gap in mobile phone ownership—with 92% of men and 91% of women now owning a phone—a substantial 32% gender gap persists in mobile internet adoption, with only 35% of women using mobile internet compared to 50% of men (GSM Association, 2024). This suggests that although mobile phone ownership is no longer a barrier for most rural women in Kenya, gaps in digital skills, affordability, and connectivity continue to constrain their full participation in the digital financial ecosystem. At the global level, digital payment solutions have played a critical role in promoting financial inclusion by enabling women to access affordable, secure, and convenient financial services using mobile devices (World Bank, 2020). According to Demirgüç-Kunt et al. (2022), the adoption of mobile money and digital payment platforms has significantly reduced gender disparities in financial account ownership and usage in developing economies. Digital payment systems have enabled women in rural areas to receive remittances, make payments, save money, and participate more effectively in economic activities without necessarily owning a conventional bank account.

In Sub-Saharan Africa, digital payment solutions have transformed the financial services landscape through the rapid expansion of mobile money platforms. Studies have shown that mobile payment technologies provide culturally acceptable and convenient alternatives to traditional banking services, thereby increasing financial participation among rural women (Bongomin et al., 2018). In one of the most comprehensive empirical studies on the subject, Suri & Jack (2016) analysed the long-term effects of mobile money adoption in Kenya using a 16-year panel dataset and found that access to mobile money significantly increased household consumption and reduced the number of households living in extreme poverty, with the impacts being more pronounced for female-headed households. These findings demonstrate that digital payment platforms not only facilitate financial transactions but also generate tangible improvements in household welfare and gender equity, underscoring their potential to drive inclusive economic development in rural communities.

In Kenya, digital payment solutions have substantially contributed to financial inclusion through innovations such as M-Pesa, Airtel Money, and Equitel. Kenya is globally recognized as a leader in mobile money innovation. The 2024 FinAccess Household Survey, conducted by the Kenya National Bureau of Statistics in collaboration with the Central Bank of Kenya and Financial Sector Deepening (FSD) Kenya, reported that formal financial access in Kenya reached 84.8% in 2024, up from 83.7% in 2021, driven largely by the increased adoption of digital payment services (Kenya National Bureau of Statistics, 2024). Mobile money uptake has been the main driver of this expansion and a key factor in narrowing the gender gap in financial inclusion to 1.6%. These platforms have enabled millions of Kenyans to conduct financial transactions conveniently and securely without visiting physical banking institutions. Even so, gains in access have not been matched by equivalent gains in financial wellbeing: Cook (2025) reports that although Kenya's formal financial inclusion gender gap narrowed to just 1.6% in 2024, the country's broader financial health gender gap actually widened to 7.5% over the same period, suggesting that digital access alone has not closed deeper disparities in women's financial outcomes.

Despite these achievements, rural women continue to experience challenges in accessing and utilizing digital payment services effectively. Factors such as limited digital literacy, inadequate digital infrastructure, low-income levels, poor network connectivity, and cultural barriers continue to hinder the full realization of financial inclusion among women in rural communities (Mugo & Kilonzo, 2017). Although mobile money services are widely available, many rural women primarily utilize them for basic transactions and have limited participation in broader financial activities that promote economic empowerment.

Beyond household-level access to finance, digital payment solutions have direct implications for women's entrepreneurial activity, which is central to the scope of this journal. Mobile money and digital wallets allow rural women running small businesses to receive customer payments instantly, separate business cash flow from household cash, and build a transaction record that can support credit applications and basic financial planning. By lowering the cost and risk of handling cash, digital payment platforms also make it easier for women traders, farmers, and small-scale entrepreneurs to reinvest working capital, pay suppliers remotely, and participate in local and regional markets. Understanding how these tools shape rural women's financial management and participation in entrepreneurial activity therefore speaks directly to ongoing debates in fintech adoption and women's economic participation.

Makueni County, located in Kenya's lower eastern region, presents a suitable context for examining the relationship between digital payment solutions and financial inclusion. The county is predominantly rural, characterized by low-income households and limited access to conventional banking services (Kenya National Bureau of Statistics, 2025). While mobile payment platforms have expanded financial service delivery within the county, rural women continue to face barriers that limit their effective use of these technologies. Challenges such as affordability of mobile devices, inadequate digital skills, and limited access to reliable network services constrain their participation in formal financial systems.

Given the growing importance of digital payment technologies in promoting inclusive finance, there is a need for empirical evidence on the relationship between digital payment solutions and financial inclusion among rural women in Makueni County. This study therefore seeks to examine the relationship between digital payment solutions and financial inclusion among rural women and to provide evidence-based recommendations that can inform policies and interventions aimed at enhancing equitable access to digital financial services in rural Kenya.

Given the gaps identified in the literature, this study examines the relationship between digital payment solutions and financial inclusion among rural women in Makueni County, Kenya. Accordingly, the following hypothesis is proposed:

H1: Digital payment solutions are not significantly associated with financial inclusion among rural women in Makueni County.

2. Review of Related Literature

2.1 Theoretical Foundations

This study draws on three complementary theoretical perspectives. The Financial Intermediation Theory is used to explain why digital platforms function as low-cost intermediaries that connect rural women to formal financial systems in place of physical bank branches. The Technology Acceptance Model (TAM) is used to explain individual-level adoption decisions, specifically how perceived usefulness and ease of use shape rural women's willingness to use digital payment platforms. The Diffusion of Innovation Theory is used to explain the pattern and pace at which digital payment innovations spread within rural communities, based on attributes such as relative advantage, compatibility, and simplicity. Together, these theories inform the study's literature review, conceptual framework, and interpretation of findings.

The study was anchored on Financial Intermediation Theory, which argues that financial intermediaries that exist within the financial systems play an important role in bringing down the transaction costs and, in the process, bridge the gap between savers and borrowers and facilitate growth. Banks have traditionally played the role of these intermediaries but have largely established presence in urban centers while ignoring the rural areas due to the absence of physical banking infrastructure. To bridge this gap, mobile platforms such as mobile money and agency banking services have emerged as modern substitutes. From a financial inclusion perspective, this theory postulates that access to digital services is more than just technological availability but goes beyond creating efficient channels that reduce such barriers, especially for marginalized groups, communities living in remote areas and women in rural areas.

Financial Intermediation Theory emphasizes structural efficiency, indicating that low transaction costs and improved accessibility enable digital platforms to allow rural women to save money securely and access credit while engaging in productive investments without having to travel long distances to physical banks. This objective is in line with the mobile money revolution in Kenya which has provided platforms such as M-Pesa that have become critical intermediaries for millions of unbanked populations across the country (Amugune et al., 2025). This theory is also limited in that as much as intermediaries reduce the costs, they actually do not automatically address other socio-cultural barriers or capability gaps. Factors such as digital literacy, affordability and trust remain substantial obstacles among rural women. Consequently, as much as financial intermediation through digital channels is necessary, it is not enough in the absence of complementary interventions that empowers women to effectively use these services.

As much as financial intermediation theory provides some important foundation for understanding the role played by financial institutions and other digital finance platforms in promoting inclusion, it is limited in its explanation of financial inclusion in the era of technology enabled services. In this study, presence of digital monetary platforms provides an avenue through which rural women in Makueni County are able to transact and access financial services. The theory focuses on the traditional banking with limited treatment of digital innovation, insufficient dynamics in gender and narrow institutional applicability and financial inclusion for women in rural areas. Additionally, informal mechanisms such as rotating savings groups and community groups may equally play a significant financial role. This theory centers on formal institutions, and therefore, it does not adequately address the role played by informal or hybrid systems, which are usually more accessible to rural women.

2.2 Empirical Review of Digital Payments and Financial Inclusion

Globally, empirical evidence has shown that digital payment solutions have greatly improved account ownership among women, although rural women remain disadvantaged in many countries (Morgal et al., 2025). In Asia, Aziz et al. (2022) explored the role of digital payment solutions on financial inclusion among rural populations in eight Asian countries, namely Bangladesh, India, Pakistan, Sri Lanka, Afghanistan, Maldives, Nepal and Bhutan. The study applied multi-level modelling methodology to get an estimate of the role of socio-economic climate on the financial inclusion of women, while taking individual features as control variables, and applied a logistic regression model. The study also used secondary data collected between 2014 and 2017 and established that there were strong correlations between the presence of technology enabled payment solutions and financial inclusion among the rural population. The research generalized the rural population, including both genders, and did not take into consideration unique socio-cultural factors biased towards women. This study will therefore address this gap by specifically looking at the rural women.

In China, Cheng et al. (2024) conducted a research study to establish the impact of digital payment solutions on the family status of women in their respective families. The study targeted rural women in Gansu Province in Western China and adopted a cross-sectional design and utilized secondary data. The findings showed that digital payment solutions promote development among women in rural areas and improve their family status. Additionally, the study noted that based on fintech, financial literacy, education, digital payment, and financial management based on fintech have a partial contribution to the family status of women, while formal savings among individuals depend on how individual women use them. This study applied secondary data, whereas this research will collect primary data from the respondents. Whereas this study includes other variables such as fintech, financial literacy, education and digital payments, this research focuses on digital payment solutions, digital financing, digital advisory and digital innovation.

In Sub-Saharan Africa, mobile money and digital channels have increased as major drivers of financial inclusion, raising financial inclusion in the region to more than double between 2014 and 2020 (Klapper, 2024). Empirical research studies have shown that digital payment solutions have a significant influence on inclusion outcomes (Amugune et al., 2025). For example, in Nigeria, a study by Okolo-Obasi et al. (2024) investigated the impact of digital payment solutions on poverty reduction among rural women in Nigeria. The study also applied a prevalence study design and used secondary information obtained between 2001 and 2022. The findings from multiple regression analysis show that technology enabled payment solutions, such as E-banking and non-interest banking strongly contribute to equitable financial participation and have played a pivotal role in reducing poverty in Nigeria by lowering transaction costs. The study further noted that direct deposits with commercial banks show a strong association with inclusive finance, although with no significant impact. Whereas the reviewed study focused on poverty reduction, the present study focuses on financial inclusion, which is not necessarily the same as poverty reduction. Additionally, the reviewed study relied on secondary data, whereas the present study relies on primary data.

Another research study by Jama et al. (2025) examined the determinants of financial inclusion in sub-Saharan Africa and applied borrowed, formal savings, and bank account ownership as key indicators. The study targeted 36,062 respondents using the 2022 World Bank Global Findex dataset and used a multivariate probit model for analysis. The findings from the study indicate that 42.2% of the urban dwellers have a formal financial account, while only 31.2% in the rural areas have such a financial account. The findings revealed that financial inclusivity is higher in urban centers than in rural areas. However, the research addressed financial inclusion in the general urban and rural population but did not specifically look at the level of inclusion of rural women. Additionally, the study used determinants of financial inclusion as measurable variables, while the current study will address the relationship linking digital payment solutions as a factor of technology enabled finance access and inclusive finance.

Kenya is regarded as one of the countries with the highest financial inclusion rates in Africa, yet rural women still continue to be less fully included compared to the national average and their counterparts in urban centers. Amugune et al. (2025) conducted a research study on the perspective of financial inclusion among rural women in Kenya. The findings indicated that the level of financial participation for women in rural areas remains relatively low, with actual engagement with diverse financial services beyond basic account ownership remaining constrained. This study assessed the perspectives of financial inclusion but left out the extent to which the presence of digital payment solutions influences financial inclusion. Additionally, the study applied one data collection instrument, which is the structured questionnaires. This research will apply a semi-structured questionnaire and key informant interviews to get in-depth data on the relationship connecting technology enabled payment solutions and access to finance for rural women. The recent study also bridges this gap by assessing the role of technology enabled payment solutions on inclusive finance.

In a related qualitative study, Wainaina et al. (2024) used an ethnographic approach to examine how rural Kenyan women's access to digital financial services shaped their economic empowerment and psychosocial wellbeing, and found that meaningful empowerment occurred only when digital financial products were designed around women's existing money-management practices and household bargaining position, while poorly fitting products produced limited or fragile gains. This qualitative evidence complements the largely quantitative studies reviewed above by showing that rural women's engagement with digital payment platforms is shaped by household and socio-cultural dynamics that structured surveys may not fully capture; the current study addresses this gap by combining quantitative and qualitative methods to capture both the extent and the lived context of rural women's engagement with digital payment solutions.

Similarly, a study by Mwaro et al. (2025) aimed at investigating the extent to which knowledge and understanding of financial concepts such as budgeting, saving, borrowing and digital banking influences the financial ability of women in Isiolo County. The research employed a depictive research design to study 279 women in business, randomly selected from a total cluster of 920 in Isiolo County. Data were gathered through structured questionnaires, and the results showed a positive correlation between financial literacy and universal financial access among the participants. However, the study did not provide the level of broad-based financial access among the target population. Additionally, the study specifically targeted women entrepreneurs both in rural and urban centers and left out other categories; hence, there is a gap to be filled by this research.

2.3 Research Gap

The reviewed studies generally agree that digital payment solutions positively contribute to equitable financial participation by improving access to monetary services, reducing transaction costs, and facilitating financial participation among rural populations. For example, Aziz et al. (2022) in Asia, Okolo-Obasi et al. (2024) in Nigeria, and Amugune et al. (2025) in Kenya discovered that digital payment platforms are positively associated with financial inclusion. In addition, Cheng et al. (2024) have shown that digital payment platforms help women to elevate their socio-economic standing through improved financial management skills.

Nevertheless, there are certain differences between the reviewed literature pieces in terms of focus, methodology, and geographical location. While Aziz et al. (2022) conducted their research on the general rural population in eight countries of Asia, Cheng et al. (2024) focused on rural women in China. In turn, Okolo-Obasi et al. (2024) focused on poverty reduction as the dependent variable rather than on financial inclusion, and Amugune et al. (2025) considered people's opinions about financial inclusion without researching the effect of digital payment platforms. As for methodology, most of the studies use secondary data and a quantitative approach, and the current research uses a mixed-method approach

3. Conceptual Framework

Figure 1 illustrates the conceptual framework guiding this study, depicting the hypothesized relationship between digital payment solutions and financial inclusion among rural women in Makueni County.

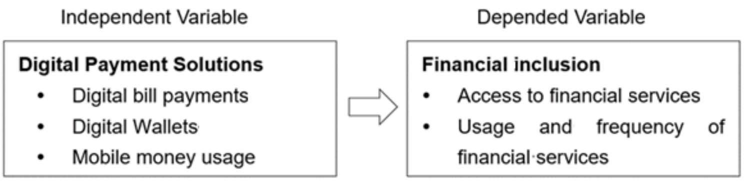

Figure 1. Interactions between digital payment solutions and financial inclusion

4. Methodology

This section presents the research methodology adopted in examining the relationship between digital payment solutions and financial inclusion among rural women in Makueni County, Kenya. It outlines the research design, target population, sampling procedures, data collection instruments, validity and reliability of research instruments, pilot study, data analysis procedures, diagnostic tests, and ethical considerations. The study adopted a mixed-methods research design that integrates both quantitative and qualitative approaches to provide a comprehensive understanding of the research problem. The mixed-methods approach was considered appropriate because it combines statistical evidence with contextual explanations regarding the use of digital payment solutions among rural women. Quantitative data enabled the researcher to establish the relationship between digital payment solutions and financial inclusion, while qualitative data provided insights into the experiences and challenges faced by rural women in accessing digital payment services. A descriptive and explanatory cross-sectional survey design was employed to collect quantitative data at a single point in time. The design enabled the study to examine the extent to which digital payment solutions are associated with financial inclusion among rural women in Makueni County. Qualitative data were collected through key informant interviews to complement and validate the quantitative findings.

The target population comprised rural women aged eighteen years and above residing in Makueni County and digital financial service agents operating within the county. Rural women were selected because they are active participants in household and income-generating activities and are among the primary beneficiaries of digital payment solutions. According to the Kenya National Bureau of Statistics (2025), Makueni County has approximately 130,154 rural women aged eighteen years and above. Additionally, the county has approximately 2,843 registered digital financial service agents who facilitate mobile money transactions and other digital payment services. These agents were included to provide qualitative insights into the adoption and utilization of digital payment solutions among rural women.

The unit of analysis for this study was the individual rural woman residing in Makueni County. The study focused on the relationship between digital payment solutions and her access to and utilization of formal financial services. The units of observation included rural women and selected digital financial service agents. Data were collected directly from these respondents using structured questionnaires and key informant interviews. The study employed Krejcie & Morgan (1970) sample size determination table to establish the sample size for rural women. From a target population of 130,154 rural women, a sample size of 384 respondents was considered adequate. Purposive sampling was used to select thirty digital financial service agents. The agents were selected based on their operational experience in serving rural communities and their involvement in digital payment transactions. The selected agents had operated for at least two years and possessed adequate knowledge regarding the use of digital payment services by rural women. The total sample size for the study was therefore 414 respondents.

The study utilized both quantitative and qualitative data collection instruments. Structured questionnaires were administered to rural women in Makueni County. The questionnaires collected information on demographic characteristics, utilization of digital payment solutions, and levels of financial inclusion. The instrument comprised closed-ended questions measured using a five-point Likert scale. The use of questionnaires enabled the researcher to collect standardized responses from a relatively large sample within a short period. Key informant interviews were conducted among selected digital financial service agents. Semi-structured interview guides were used to collect qualitative information. The interviews enriched the quantitative findings by providing practical experiences and contextual explanations of digital payment practices in rural areas. Validity and reliability were considered essential in ensuring the credibility and accuracy of the research findings.

Content validity was achieved through an extensive review of literature on digital payment solutions and financial inclusion. The questionnaires were reviewed by supervisors and experts in finance and research methodology to ensure that the items adequately captured the study constructs. Construct validity was enhanced through the operationalization of variables based on established indicators commonly applied in financial inclusion studies. The use of multiple data collection methods further strengthened the validity of the study through methodological triangulation. Reliability was assessed using Cronbach's alpha coefficient to determine the internal consistency of the questionnaire items. A pilot study was conducted among rural women in Machakos County, which shares similar socio-economic characteristics with Makueni County. Cronbach's alpha values of 0.70 and above were considered acceptable for all constructs.

The reliability coefficients presented in Table 1 indicate that the research instrument was internally consistent and suitable for data collection.

Table 1. Reliability test results

Variable	Number of Items	Cronbach's alpha	Interpretation
Financial inclusion	9	0.842	Reliable

A pilot study was conducted using approximately ten percent of the intended sample size among rural women in Machakos County. The pilot study assessed the clarity, relevance, and reliability of the questionnaire items. Feedback obtained during the pilot study informed the revision of ambiguous questions and improved the overall quality of the research instrument. Quantitative data collected through questionnaires were coded and entered into the SPSS (Statistical Package for Social Sciences) for analysis. Data cleaning procedures were undertaken to identify missing values and inconsistencies before analysis. Descriptive statistics were used to summarize the characteristics of respondents and study variables. Inferential statistics were employed to establish the relationship between digital payment solutions and financial inclusion. Qualitative data obtained through key informant interviews were analyzed using thematic analysis. Responses were transcribed, coded, and categorized into themes relevant to the study objectives.

The study obtained ethical clearance from the Catholic University of Eastern Africa and a research permit from the NACOSTI (National Commission for Science, Technology and Innovation) before commencing data collection. Participation in the study was voluntary, and informed consent was obtained from all respondents. Confidentiality and anonymity were assured throughout the study. Participants were informed that the information provided would be used strictly for academic purposes. Data collected were securely stored and will be destroyed after the stipulated retention period in accordance with institutional ethical guidelines. The researcher also ensured that respondents were not exposed to any form of harm, coercion, or discrimination during the data collection process.

5. Findings

372 of the 384 questionnaires that were sent to rural women in the study's sample were successfully completed and returned, yielding a 96.9% questionnaire response rate. 12 questionnaires were not included in the analysis because they were either incomplete or not returned. The study's qualitative component had a 100% response rate since all 30 of the purposively chosen key informants took part in the interviews.

5.1 Qualitative Findings From Key Informant Interviews

Responses from the 30 key informant interviews with digital financial service agents were analyzed thematically alongside the quantitative results. Four broad themes emerged as recurring across the interviews: network and connectivity access, affordability of service and transaction charges, fraud and security concerns, and rural women's digital skills.

Network and connectivity access. Agents consistently described unreliable mobile network coverage in parts of Makueni County as a practical constraint on digital payment use, particularly in more remote sub-locations, where transactions are sometimes delayed or fail during periods of poor signal.

Affordability of service charges. Several agents noted that transaction fees and withdrawal charges remain a source of hesitation for some rural women, especially for frequent, low-value transactions, even where the underlying platforms are otherwise seen as convenient.

Fraud and security concerns. Agents raised concerns about fraud, including impersonation and SIM-swap related scams targeting less experienced users, and noted that such incidents can undermine trust in digital platforms even when they are not widespread.

Digital skills. Agents observed that women with limited digital literacy tend to rely on agents for tasks that more experienced users perform independently, and that basic training or hands-on support at the point of registration meaningfully improves independent use over time.

5.2 Descriptive Statistics for Digital Payment Solutions

The evaluation of digital payment solutions involved eight statements focusing on how accessible, affordable, convenient, secure, and efficient these platforms are for rural women.

The findings presented in Table 2 indicate that respondents generally agreed that the use of digital payment solutions has significantly improved their financial transactions and enhanced their participation in formal financial systems. The high mean scores across the various indicators demonstrate that digital payment platforms are widely accepted and positively perceived by rural women in Makueni County. This suggests that digital payment technologies have become important tools for promoting financial inclusion by providing accessible, convenient, and cost-effective financial services.

Table 2. Descriptive statistics for digital payment solutions (n = 372)

Statement	Mean	Std. Dev.
Mobile money services are easily accessible in my locality.	4.28	0.71
I regularly use mobile money services to receive payments.	4.17	0.76
Digital payment platforms have reduced the cost of financial transactions.	4.05	0.83
Digital payment services are convenient for my daily financial activities.	4.22	0.69
I trust the security of digital payment platforms.	3.98	0.87
Digital payment services have reduced the need to travel to banking institutions.	4.31	0.65
Digital payment platforms have improved my ability to save money.	4.09	0.78
Digital payment services are easy to learn and use.	4.25	0.73
Composite mean	4.17	0.75

Note: n = number of respondents. All items were measured on a five-point Likert scale. Std. Dev. = standard deviation.

The highest-rated statement was that digital payment services have reduced the need for physical visits to banking institutions, recording a mean score of 4.31 and a standard deviation (SD) of 0.65. This finding implies that rural women can conveniently perform financial transactions from their localities without incurring additional transportation costs or spending considerable time travelling to banks. The reduction in physical barriers to financial services is particularly important in rural settings where banking infrastructure may be limited or located far from residential areas. Digital payment platforms have therefore contributed significantly to improving access to financial services among rural women.

The second highest-rated item showed that mobile money services are easily accessible, with a mean score of 4.28 (SD = 0.71). This finding demonstrates that digital payment services are readily available within rural communities and are sufficiently accessible to facilitate everyday financial transactions. Accessibility is a key component of financial inclusion since individuals can only utilize financial services when they are available and easy to access.

Additionally, respondents agreed that digital payment services are convenient and user-friendly, recording a mean score of 4.25 (SD = 0.73). This suggests that rural women are comfortable using digital payment platforms for their financial needs. The simplicity and convenience associated with mobile money services have encouraged greater adoption of digital financial technologies among the respondents.

Furthermore, the study established that digital payment solutions have the strongest positive correlation with financial inclusion, with a Pearson correlation coefficient (r) of 0.731 and a probability value (p) of less than 0.01. This strong and statistically significant relationship implies that improvements in accessibility, convenience, affordability, and reliability of digital payment systems substantially enhance the level of financial inclusion among rural women. Respondents who frequently utilize mobile money and other digital payment platforms are more likely to access formal financial services, save money digitally, receive payments conveniently, and improve their overall financial well-being.

The findings therefore demonstrate that digital payment solutions are fundamental drivers of financial inclusion in Makueni County. By reducing geographical and financial barriers and providing user-friendly financial services, digital payment platforms continue to empower rural women economically and increase their participation in the formal financial sector.

The statistics from Table 3 depict a multiple correlation coefficient ($R = 0.842$), indicating a strong positive relationship between the predictor variables and financial inclusion. The regression model included Digital Payment Solutions, Digital Financing, and Digital Advisory Services as predictors of Financial Inclusion. The coefficient of determination ($R^2 = 0.709$) implies that 70.9% of the variance in financial inclusion of rural women can be attributed jointly to these three predictors, with digital payment solutions being the focus of this article.

Table 3. Model summary

Model	***R***	***R*²**	**Adjusted *R*²**	**Std. Error of Estimate**
1	0.842	0.709	0.706	0.386

Note: R = multiple correlation coefficient; Adjusted R^2 = adjusted coefficient of determination; Std. Error = standard error.

To ascertain whether the whole regression model was statistically significant in explaining financial inclusion, the analysis of variance (ANOVA) test was used.

The ANOVA results presented in Table 4 revealed that the regression equation is statistically significant, with an F-statistic (F) of 299.524 ($p < 0.001$). Since the significance level is less than 0.05, the null hypothesis stating that there is no statistical significance in the regression equation was rejected. This finding confirms that the three-predictor model (Digital Payment Solutions, Digital Financing, and Digital Advisory Services) is appropriate for explaining financial inclusion among rural women in Makueni County. The statistical significance of the model demonstrates that digital payment solutions, alongside the other two predictors, play an important role in enhancing financial inclusion.

Table 4. Analysis of variance results

Model	**Sum of Squares**	***df***	**Mean Square**	***F***	**Sig.**
Regression	133.974	3	44.658	299.524	< 0.001

Note: df = degrees of freedom; Sig. = significance.

Table 5 presents the regression coefficients for the three predictor variables.

Table 5. Regression coefficients

Variable	**Unstandardized *B***	**Std. Error**	**Standardized *β***	***t*-Value**	**Sig.**
Constant	0.548	0.171	-	3.204	0.001
Digital payment solutions	0.382	0.048	0.398	7.958	< 0.001
Digital financing	0.296	0.045	0.307	6.578	< 0.001
Digital advisory services	0.241	0.043	0.259	5.605	< 0.001

Note: B = unstandardized regression coefficient; β = standardized regression coefficien; Std. Error = standard error; Sig. = significance. Hyphen (-) indicates not applicable because the constant term has no standardized coefficient.

5.3 Dependent Variable: Financial Inclusion

The study sought to examine the relationship between digital payment solutions and financial inclusion among rural women in Makueni County. The descriptive findings indicated that respondents generally agreed that digital payment solutions are accessible, convenient, secure, and user-friendly. Digital payment solutions recorded a composite mean score of 4.17, demonstrating a high level of acceptance and utilization among the respondents. The findings suggest that rural women have increasingly embraced digital payment platforms as essential tools for managing their financial transactions and accessing financial services.

The high composite mean score further implies that digital payment solutions have successfully addressed many of the traditional barriers associated with financial inclusion in rural settings. Through mobile money services and other digital payment platforms, rural women are able to conduct transactions conveniently without necessarily visiting banking institutions. The availability and ease of using these technologies have significantly contributed to improving financial accessibility and promoting financial participation among women in rural communities.

The correlation analysis established that there is a strong positive and statistically significant relationship between digital payment solutions and financial inclusion ($r = 0.731$, $p < 0.01$). This finding indicates that improvements in accessibility, convenience, affordability, and reliability of digital payment services are associated with higher levels of financial inclusion. Rural women who actively utilize digital payment platforms are more likely to access formal financial services, save money digitally, receive payments efficiently, and improve their financial well-being.

Furthermore, regression analysis demonstrated that digital payment solutions have a positive and statistically significant association with financial inclusion, with an unstandardized regression coefficient (B) of 0.382 ($p < 0.001$). The positive regression coefficient implies that higher utilization and accessibility of digital payment solutions is linked to a corresponding increase in financial inclusion among rural women. The findings therefore provide empirical evidence that digital payment solutions are associated with enhanced financial inclusion in Makueni County and should continue to be promoted as an effective mechanism for improving access to formal financial services among rural women.

6. Discussions

The first research objective sought to examine the relationship between digital payment solutions and financial inclusion among rural women in Makueni County, Kenya. The study findings indicate that digital payment solutions are positively associated with financial inclusion by improving access to formal financial services and enabling rural women to participate effectively in the digital financial ecosystem. The findings demonstrate that increased accessibility, affordability, convenience, and ease of use of digital payment platforms significantly contribute to improving financial inclusion among rural women.

The descriptive statistics revealed a high level of acceptance and utilization of digital payment solutions among the respondents. The composite mean score of 4.17 (SD = 0.75) indicates that respondents generally agreed that digital payment solutions are accessible, convenient, secure, and beneficial in facilitating their financial activities. These findings suggest that rural women in Makueni County have increasingly embraced digital payment technologies as viable alternatives to conventional banking services.

Among the various indicators of digital payment solutions, the highest-rated statement was that digital payment services have reduced the need to travel to banking institutions, which recorded a mean score of 4.31 (SD = 0.65). This finding is particularly significant given the geographical and infrastructural challenges that often characterize rural areas. Digital payment platforms have enabled rural women to access financial services remotely through their mobile phones, thereby reducing transportation costs and the time previously spent travelling to financial institutions. Consequently, digital payment solutions appear to have contributed to bridging the financial access gap that has historically excluded many rural populations from formal financial systems.

The findings further revealed that mobile money services are easily accessible within respondents' localities, as reflected by a mean score of 4.28 (SD = 0.71). Accessibility remains one of the most important determinants of financial inclusion since financial services can only be utilized when they are readily available to users. The widespread availability of mobile money agents and digital payment infrastructure in rural communities has significantly enhanced rural women's ability to transact financially whenever the need arises. Increased accessibility has consequently promoted greater participation in formal financial systems among women who would otherwise remain financially excluded.

Additionally, respondents indicated that digital payment services are easy to learn and use, with a mean score of 4.25 (SD = 0.73). Ease of use is a critical determinant of technology adoption, particularly among rural populations that may have varying levels of digital literacy. The simplicity of mobile payment platforms appears to have encouraged greater adoption among rural women by reducing technological complexities that often discourage the use of financial technologies. This implies that user-friendly digital financial services are instrumental in advancing financial inclusion initiatives in rural communities.

Similarly, respondents agreed that digital payment services are convenient for their daily financial activities, as evidenced by a mean score of 4.22 (SD = 0.69). Convenience in financial transactions significantly influences individuals' willingness to adopt and continuously utilize financial technologies. The ability to send and receive money instantly, pay for goods and services electronically, and access financial services at any time has transformed the manner in which rural women manage their personal and business finances.

Furthermore, the study established that digital payment platforms have improved respondents' saving behavior, recording a mean score of 4.09 (SD = 0.78). This finding suggests that digital payment solutions not only facilitate transactions but also encourage better financial management practices among rural women. By enabling users to store money electronically and access various savings products, digital payment platforms contribute to improved financial discipline and economic empowerment.

The findings on financial inclusion further support the significant role played by digital payment solutions. The composite mean score for financial inclusion was 4.15 (SD = 0.73), indicating that respondents perceived themselves as highly financially included. The highest-rated indicator of financial inclusion was the ability to send and receive money digitally, which recorded a mean score of 4.35 (SD = 0.63). This demonstrates that digital payment solutions have substantially improved financial accessibility and participation among rural women.

The respondents further agreed that digital financial services have improved their overall financial well-being ($M = 4.20$, SD = 0.70) and increased access to formal financial services ($M = 4.18$, SD = 0.73). Improved financial well-being is a key outcome of financial inclusion because it reflects the extent to which individuals are able to access and effectively utilize financial services to improve their socio-economic conditions. The findings therefore suggest that digital payment solutions have positively influenced both financial access and financial outcomes among rural women in Makueni County.

The inferential statistical analysis provides further empirical support for the relationship between digital payment solutions and financial inclusion. Pearson's Product Moment Correlation analysis established a strong positive and statistically significant relationship between digital payment solutions and financial inclusion ($r = 0.731$, $p < 0.01$). This finding implies that improvements in digital payment solutions are associated with increased levels of financial inclusion among rural women.

The strength of the correlation coefficient indicates that digital payment solutions are important correlates of financial inclusion. Rural women who frequently utilize mobile money services and other digital payment platforms are more likely to access formal financial services, save money digitally, improve their household financial management practices, and enhance their overall financial well-being. The findings therefore confirm that digital payment technologies provide practical solutions to many of the financial barriers experienced by rural women.

Regression analysis further established that digital payment solutions have a positive and statistically significant association with financial inclusion, alongside digital financing and digital advisory services in the same model. The regression coefficient for digital payment solutions was positive and significant ($B = 0.382$, $t = 7.958$, $p < 0.001$), the largest of the three predictors, indicating that higher digital payment solutions scores are linked to a corresponding increase in financial inclusion among rural women, holding the other two predictors constant. The statistical significance of the findings provides sufficient evidence to reject the null hypothesis that digital payment solutions have no significant association with financial inclusion.

Moreover, the model summary revealed that digital payment solutions, digital financing, and digital advisory services jointly explain approximately 70.9% ($R^2 = 0.709$) of the variation in financial inclusion among rural women in Makueni County, with digital payment solutions contributing the largest individual share among the three. This finding demonstrates that digital payment solutions are a major predictor of financial inclusion within the study context. Although other factors may also influence financial inclusion, digital payment technologies account for a substantial proportion of the observed variation in financial inclusion outcomes.

The findings of this study are consistent with the TAM, which posits that individuals are more likely to adopt technologies that they perceive as useful and easy to use. The high levels of agreement among respondents regarding the accessibility, convenience, and user-friendliness of digital payment services provide empirical support for this theory. Rural women appear willing to adopt digital payment technologies because they effectively address their financial needs while remaining relatively simple to operate.

The study findings also support the propositions of the Diffusion of Innovation Theory, which argues that innovations are adopted when they provide clear advantages over existing practices. Digital payment solutions have demonstrated several relative advantages over traditional banking systems, including convenience, accessibility, affordability, and efficiency. The rapid adoption of mobile money services among rural women suggests that digital payment platforms possess characteristics that encourage widespread acceptance within rural communities.

Furthermore, the findings are consistent with previous empirical studies discussed in Section 3, which found that digital payment mechanisms are positively associated with financial inclusion by reducing geographical and financial barriers to accessing financial services. Digital payment platforms have enabled marginalized populations, particularly women in rural areas, to participate more actively in formal financial systems without necessarily relying on conventional banking infrastructure.

The present study contributes to the existing body of literature by demonstrating that digital payment solutions remain one of the most influential dimensions of digital financial services in promoting financial inclusion among rural women. By improving accessibility, reducing transaction costs, facilitating digital savings, and enhancing financial convenience, digital payment solutions have significantly transformed the financial lives of rural women in Makueni County.

In conclusion, the findings of this study provide empirical evidence that digital payment solutions are strongly associated with financial inclusion among rural women. The positive descriptive, correlation, and regression results collectively demonstrate that mobile money services and other digital payment platforms are linked to more equitable access to financial services. The rejection of the null hypothesis confirms that digital payment solutions are among the strongest correlates of financial inclusion and should therefore remain central to policy interventions and financial sector innovations aimed at empowering rural women economically.

7. Conclusions

The study concludes that digital payment solutions are strongly linked to financial inclusion among rural women in Makueni County, Kenya. The findings demonstrate that mobile money and other digital payment platforms are associated with improved access to formal financial services through convenient, affordable, and secure mechanisms for conducting financial transactions. Digital payment solutions are associated with a reduction in traditional barriers linked to financial inclusion, including long distances to banking institutions, high transaction costs, and limited access to formal financial systems.

The descriptive findings revealed that respondents had a high level of agreement regarding the benefits of digital payment solutions, with a composite mean score of 4.17 (SD = 0.75). The highest-rated statement indicated that digital payment services have reduced the need to travel to banking institutions (M = 4.31, SD = 0.65), while mobile money services were found to be highly accessible within local communities (M = 4.28, SD = 0.71). Respondents further agreed that digital payment platforms are easy to learn and use (M = 4.25, SD = 0.73), convenient for daily financial activities (M = 4.22, SD = 0.69), and contribute positively to their saving behaviour (M = 4.09, SD = 0.78).

Similarly, the findings on financial inclusion showed a high composite mean score of 4.15 (SD = 0.73), indicating substantial levels of financial inclusion among rural women. The ability to send and receive money digitally recorded the highest mean score (M = 4.35, SD = 0.63), demonstrating the critical role of digital payment solutions in facilitating financial transactions. Digital financial services were also found to improve respondents' overall financial well-being (M = 4.20, SD = 0.70) and increase access to formal financial services (M = 4.18, SD = 0.73).

The inferential statistical findings further support the importance of digital payment solutions in promoting financial inclusion. Pearson's correlation analysis revealed a strong positive and statistically significant relationship between digital payment solutions and financial inclusion ($r = 0.731$, $p < 0.01$). This finding implies that increased accessibility, convenience, and reliability of digital payment platforms are associated with higher levels of financial inclusion among rural women in Makueni County.

Moreover, regression analysis established that digital payment solutions, alongside Digital Financing and Digital Advisory Services, have a positive and statistically significant association with financial inclusion ($B = 0.382$, $t = 7.958$, $p < 0.001$ for digital payment solutions, the strongest of the three predictors). The full model explained 70.9% ($R^2 = 0.709$) of the variation in financial inclusion, indicating that digital payment solutions are among the strongest predictors of rural women's participation in formal financial systems. The study therefore concludes that continued investment in accessible and user-friendly digital payment platforms has significant potential to strengthen financial inclusion and support the socio-economic well-being of rural women in Makueni County.

8. Limitations

This study is subject to some limitations. The cross-sectional survey design captures associations at a single point in time and does not establish causal relationships between digital payment solutions and financial inclusion; the language used to describe the findings throughout this article should therefore be read as describing relationships and associations rather than proven causal effects. In addition, the data were collected from rural women and digital financial service agents in a single county, Makueni, and the findings may not generalize to rural women in other counties or ecological zones in Kenya. Future longitudinal or multi-county research would help establish whether these relationships hold over time and across different contexts.

9. Recommendations

The Government of Kenya, via its relevant ministries and regulating bodies, needs to continue to make investments in ICT (Information and Communication Technology) infrastructure, especially in rural areas, since improved connectivity, availability of power, and mobile network will improve the access of rural women to digital monetary services. Further, the government needs to come up with policies that will make sure that there are affordable digital financial services while ensuring that there is protection from cyber fraud and lending charges among others. Commercial banks, microfinance institutions, savings and credit cooperative organizations, and mobile money operators need to keep innovating low-cost, intuitive, and accessible digital finance products designed for rural women's needs. Financial institutions need to make account openings cheaper, eliminate transaction fees, provide efficient customer care services, and design customized digital saving and credit products for women running small business ventures and agriculture-based businesses.

Financial institutions, county governments, development partners, and non-governmental organizations should develop digital financial literacy programs aimed at rural women. Financial exclusion will be reduced and confidence in using digital financial services will increase with digital financial literacy. Financial institutions, mobile network operators, and tech businesses should continue to develop innovative digital financial technologies that will improve security, affordability, and accessibility. It is also important to promote affordable smartphones, enhanced internet connectivity, and improvement of mobile applications that are accessible to illiterate people. Future research may include conducting the same type of research in different counties in Kenya in order to establish whether the results are consistent throughout different geographical locations. Researchers may also conduct comparative studies on rural and urban areas in order to establish any difference in the adoption of technology based monetary services and financial inclusion. Longitudinal research may be conducted to study the relationship between digital financial services and universal financial access.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflicts of interest.

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